

1. How does sampling affect image quality? How to correct blur?

Sampling determines the no. of pixels used to represent an image. Low sampling causes blur and loss of fine details. Increasing the sampling rate (higher resolution) improves image quality. Image interpolation and proper camera settings can also help restore clarity.

2. How does image sensing and acquisition affect low-light images?

In low light, the sensor receives less light, resulting in dark and noisy images. Poor acquisition reduces image quality and detail. Using a high-quality sensor, increasing exposure time, or adding illumination improves the captured image. Image enhancement techniques can further improve visibility.

3. What causes aliasing artifacts? How to correct them?

Aliasing occurs when an image is sampled below the required sampling rate. It creates jagged edges and false patterns in the image. Increasing the sampling frequency reduces aliasing. Applying an anti-aliasing filter before sampling also helps prevent these artifacts.

4. Differentiate between pixel resolution and intensity resolution (Medical Imaging)

Pixel resolution determines the spatial detail and sharpness of an image. Intensity resolution determines the number of gray levels that can be represented. High pixel resolution helps identify small structures while high intensity resolution improves tissue contrast. Both are essential for accurate medical diagnosis.

5. Computer vision level in a real time surveillance system.

A surveillance system mainly uses high-level computer vision because it recognizes and tracks people or objects. It also analyzes activities and makes decisions such as detecting suspicious behavior. Low-level processing enhances the image before analysis. Together, these levels enable accurate real-time monitoring.

6. How do image formation models improve autonomous driving?

Image formation models describe how light interacts with the camera to create images. They help correct distortion and improve image quality. This enables accurate lane detection, obstacle recognition, and traffic



Sign Identification • Better image formation increases the safety and reliability of autonomous vehicles

7. Role of quantization in converting a grayscale image to digital.

Quantization converts continuous gray levels into discrete digital values. Higher quantization levels preserve more image details and smoother transitions. Lower quantization may cause loss of detail and banding effects. It directly affects the quality of the digital image.

8. How does acquisition contribute to image noise? How to reduce it?

Noise is introduced by image sensors, low light conditions, and electronic interference during acquisition. It reduces image clarity and accuracy. Better sensors, proper lighting, and optimized camera settings reduce noise. Image filtering techniques can further improve the final image.

9. Computer Vision levels in facial recognition.

Low-level vision captures and enhances the image. Mid-level vision detects the face and extracts important facial features. High-level vision compares these features with a database to identify the person. The final decision is based on pattern recognition and classification.

10. How do sampling and image formation affect object visibility?

low sampling reduces image resolution, making objects appear unclear or blurred. poor image formation can introduce distortion and reduce contrast. increasing the sampling rate improves object visibility. proper camera focus, lighting and image formation techniques help objects more accurately.

How to detect